

Exercise 1.

X and Y are IID $\exp(1)$

$$U = X$$

$$V = X + Y$$

$$f_{uv}(u, v) = f_{xy}(x, y)|J|$$

$$J = \left| \det \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \right|$$

$$J = |1 - 0|$$

$$J = 1$$

$$f_{uv}(u, v) = f_{xy}(x, y) \times 1, x > 0, y > 0$$

$$f_{uv}(u, v) = e^{-x}e^{-y}, x > 0, y > 0$$

$$f_{uv}(u, v) = e^{-x-y}, x > 0, y > 0$$

$$U = X$$

$$V = U + Y \rightarrow Y = V - U$$

$$f_{uv}(u, v) = e^{-v}, 0 \leq u \leq v \leq \infty$$

$$f_v(v) = \int_0^v e^{-v} du$$

$$f_v(v) = ue^{-v}|_0^v$$

$$f_v(v) = ve^{-v}, 0 \leq v$$

Exercise 2.

$$U = X$$

$$V = \frac{Y}{X}$$

$$Y = VU$$

$$J = |\det \begin{pmatrix} 1 & 0 \\ v & u \end{pmatrix}|$$

$$J = |u - 0|$$

$$J = |u| = u$$

$$f_{uv}(u, v) = f_{xy}(x, y)|u|$$

$$f_{uv}(u, v) = e^{-u-vu}u, 0 \leq u, 0 \leq v$$

$$f_v(v) = \int_0^\infty ue^{-u(1+v)} du$$

$$a = u(1 + v)$$

$$u = \frac{a}{1 + v}$$

$$du = \frac{da}{1 + v}$$

$$f_v(v) = \int_0^\infty \frac{a}{1+v} e^{-a} \times \frac{a}{1+v}$$

$$f_v(v) = \frac{1}{(1+v)^2} \int_0^\infty ae^{-a} da$$

$$f_v(v) = \frac{1}{(1+v)^2} [-ae^{-a} + \int_0^\infty e^{-a} da$$

$$f_v(v) = \frac{1}{(1+v)^2} [-ae^{-a} - e^{-a}|_0^\infty]$$

$$f_v(v) = \frac{1}{(1+v)^2} [-u(1+v)e^{-u(1+v)} - e^{-u(1+v)}|_0^\infty]$$

$$f_v(v) = \frac{1}{(1+v)^2} [e^{-0}]$$

$$f_v(v) = \frac{1}{(1+v)^2}, v \geq 0$$

Exercise 3.

$$X \sim \text{Gamma}(\alpha, \beta)$$

$$Y = \frac{1}{X}$$

$$X = \frac{1}{Y}, y > 0$$

$$f_y(y) = f_x(x) \left| \frac{dx}{dy} \right|$$

$$f_y(y) = \frac{(\frac{1}{y})^{\alpha-1} e^{-\frac{1}{y\beta}}}{\beta^\alpha \Gamma(\alpha)} \left| -\frac{1}{y^2} \right|$$

$$f_y(y) = \frac{e^{-\frac{1}{y\beta}}}{\beta^\alpha \Gamma(\alpha)} \left(\frac{1}{y}\right)^{\alpha-1} \left(\frac{1}{y}\right)^2$$

$$f_y(y) = \frac{e^{-\frac{1}{y\beta}}}{\beta^\alpha \Gamma(\alpha)} \left(\frac{1}{y}\right)^{\alpha+1}, y > 0$$

Exercise 4.

$$N \sim N(0, 1)$$

$$Y \sim X_k^2$$

$$f(x, y) = \frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}} \times \frac{y^{\frac{k}{2}-1} e^{-\frac{y}{2}}}{2^{\frac{k}{2}} \Gamma(\frac{k}{2})}$$

$$U = \frac{X}{\sqrt{\frac{Y}{k}}} \rightarrow X = U \sqrt{\frac{V}{k}}$$

$$V = Y$$

$$J = \left| \det \begin{pmatrix} \sqrt{\frac{V}{k}} & 0 \\ \frac{U\sqrt{k}}{2k} & 1 \end{pmatrix} \right|$$

$$J = \left| \sqrt{\frac{V}{k}} - 0 \right|$$

$$J = \sqrt{\frac{V}{k}}$$

$$f_{uv}(u, v) = \frac{1}{\sqrt{2\pi k} 2^{k/2} \Gamma(\frac{k}{2})} e^{-v(\frac{u^2}{2k} + \frac{1}{2})} v^{\frac{k+1}{2}-1}$$

$$f_u(u) = \int_0^\infty \frac{1}{\sqrt{2\pi k} 2^{k/2} \Gamma(\frac{k}{2})} e^{-v(\frac{u^2}{2k} + \frac{1}{2})} v^{\frac{k+1}{2}-1} dv$$

$$f_u(u) = \frac{1}{\sqrt{2\pi k} 2^{k/2} \Gamma(\frac{k}{2})} \int_0^\infty e^{-v(\frac{u^2}{2k} + \frac{1}{2})} v^{\frac{k+1}{2}-1} dv$$

$$f_u(u) = \frac{1}{\sqrt{2\pi} 2^{k/2} \Gamma(\frac{k}{2})} \int_0^\infty e^{-v(\frac{u^2+k}{2k})} v^{\frac{k+1}{2}-1} dv$$

$$f_u(u) = \frac{1}{\sqrt{2\pi} 2^{k/2} \Gamma(\frac{k}{2})} \left(\frac{2k}{u^2+k}\right)^{\frac{k+1}{2}} \Gamma\left(\frac{k+1}{2}\right)$$

$$f_u(u) = \frac{2^{k/2} \sqrt{2}}{\sqrt{2} \sqrt{\pi k} 2^{k/2} \Gamma(\frac{k}{2})} \left(\frac{k}{u^2+k}\right)^{\frac{k+1}{2}} \Gamma\left(\frac{k+1}{2}\right)$$

$$f_u(u) = \frac{\Gamma(\frac{k+1}{2})}{\sqrt{\pi k} \Gamma(\frac{k}{2})} \left(\frac{k}{u^2+k}\right)^{\frac{k+1}{2}}$$

Exercise 5.(a) Z_1, Z_2 IID standard normals

$$W = y = Z_1 + Z_2$$

$$Z_2 = W - Z_1$$

$$f_{z_1, w}(z_1, w) = f_{z_1, z_2}(z_1, w - z_1)$$

$$f_{z_1, w}(z_1, w) = f_{z_1}(z_1) f_{z_2}(w - z_1)$$

$$= \frac{1}{\sqrt{2\pi}} e^{-\frac{z_1^2}{2}} \times \frac{1}{\sqrt{2\pi}} e^{-\frac{(w-z_1)^2}{2}}$$

$$= \frac{1}{2\pi} e^{-\frac{z_1^2}{2} - \frac{(w-z_1)^2}{2}}$$

$$N(0, 2) - f_W(w) = \frac{1}{\sqrt{4\pi}} e^{-\frac{w^2}{4}}$$

$$f_{z_1|w}(z_1|w) = \frac{f_{z_1, w}(z_1, w)}{f_w(w)} = \frac{\frac{1}{2\pi} e^{-\frac{z_1^2}{2} - \frac{(w-z_1)^2}{2}}}{\frac{1}{\sqrt{4\pi}} e^{-\frac{w^2}{4}}}$$

$$= \frac{2\pi^{1/2} e^{-\frac{-w^2 + 2wz_1 - 2z_1^2}{2}}}{2\pi e^{-\frac{w^2}{4}}}$$

$$= \frac{1}{\sqrt{\pi}} e^{-w^2/4 + wz_1 - z_1^2}$$

$$= \frac{1}{\sqrt{\pi}} e^{-(z_1 - w/2)^2}$$

(b) $Z_1, Z_2 \sim N(0, 1)$

$$U = \frac{z_1}{z_2} \rightarrow z_1 = UV$$

$$V = z_2$$

$$J = |\det \begin{pmatrix} V & U \\ 0 & 1 \end{pmatrix}|$$

$$J = |V - 0| = V$$

$$f_{uv}(u, v) = f_{z_1, z_2}(x, y) |J|$$

$$f_{uv}(u, v) = \frac{v}{2\pi} e^{-\frac{(u^2 v^2 + v^2)}{2}}$$

$$f_u(u) = \int_{-\infty}^{\infty} \frac{v}{2\pi} e^{-\frac{(u^2 v^2 + v^2)}{2}} dv$$

$$f_u(u) = \frac{1}{2\pi} \int_{-\infty}^{\infty} v e^{-\frac{(u^2 v^2 + v^2)}{2}} dv$$

$$w = \frac{v^2(1 + u^2)}{2}$$

$$dw = v(1 + u^2)dv$$

$$f_u(u) = \frac{1}{\pi(1+u^2)} \int_0^{\infty} e^{-w} dw$$

$$f_u(u) = \frac{1}{\pi(1+u^2)}$$

Exercise 6.

$X_1, X_2, \dots, X_n$ are IID $\exp(\lambda)$

$$P(Y_1 > y) = P(X_1 > y, X_2 > y, \dots, X_n > y)$$

$$P(Y_1 > y) = P(X_1 > y)P(X_2 > y) \dots P(X_n > y)$$

$$P(Y_1 > y) = P(X_1 > y)P(X_1 > y) \dots P(X_1 > y)$$

$$P(Y_1 > y) = [P(X_1 > y)]^n$$

$$P(Y_1 > y) = [\lambda e^{-\lambda y}]^n = \lambda e^{-n\lambda y}, y \geq 0$$

$$F_{Y_1}(y) = 1 - P(Y_1 > y) = 1 - \lambda e^{-n\lambda y}, y \geq 0$$

$$f_{y_1}(y) = \frac{d}{dy}(1 - \lambda e^{-n\lambda y}) = n\lambda e^{-n\lambda y}, y \geq 0$$

$$Y_1 \sim \exp(n\lambda)$$

Exercise 7.

$X_1, X_2, \dots, X_n$ are independent exponential, $X_i \sim \exp(\lambda_i)$

$$X_i \sim \exp(\lambda_i) \rightarrow \lambda_i e^{\lambda_i x}$$

$$P(Y_1 > y) = P(X_1 > y, X_2 > y, \dots, X_n > y)$$

$$P(Y_1 > y) = P(X_1 > y)P(X_2 > y) \dots P(X_n > y)$$

$$P(X_i > y) = \lambda_i e^{-\lambda_i y}, y \geq 0$$

$$P(Y_1 > y) = \prod_{i=1}^n P(X_i > y) = \prod_{i=1}^n e^{-\lambda_i y} = e^{-y \sum_{i=1}^n \lambda_i}, y \geq 0$$

$$F_{Y_1}(y) = 1 - e^{-y \sum_{i=1}^n \lambda_i}, y \geq 0$$

$$f_{y_i}(y) = \frac{d}{dy}(1 - e^{-y \sum_{i=1}^n \lambda_i}), y \geq 0$$

$$f_{Y_1}(y) = \sum_{i=1}^n \lambda_i e^{-y \sum_{i=1}^n \lambda_i}, y \geq 0$$

$$Y_1 - \exp(\sum_{i=1}^n \lambda_i)$$

Exercise 8.

$$f(x) = 2x, 0 < x < 1$$

$$F(x) = x^2, 0 < x < 1$$

$$(a) f_{Y_2} = \frac{n!}{(j-1)!(n-j)!} \times (x^2)^{j-1} \times 2x \times (1-x^2)^{n-j}, 0 < y < 1$$

$$j = 2, n = 3$$

$$f_{Y_2} = \frac{3!}{(2-1)!(3-2)!} \times (x^2)^{2-1} \times 2x \times (1-x^2)^{3-2}, 0 < y < 1$$

$$f_{Y_2}(y_2) = \frac{3!}{1!1!} (x^2)^1 2x (1-x^2)^1, 0 < y < 1$$

$$f_{Y_2}(y_2) = 6 \times 2x^3 (1-x^2), 0 < y < 1$$

$$f_{Y_2}(y_2) = 12x^3 (1-x^2), 0 < y < 1$$

$$f_{Y_2}(y_2) = 12x^3 - 12x^5, 0 < y < 1$$

$$f_{Y_2}(y_2) = \begin{cases} 12y^3 - 12y^5, 0 < y < 1 \\ 0, otherwise \end{cases}$$

$$(b) E(Y_2) = \int_0^1 y(12y^3 - 12y^5) dy$$

$$E(Y_2) = \int_0^1 12y^4 - 12y^6 dy$$

$$E(Y_2) = \frac{12}{5}y^5 - \frac{12}{7}y^7 \Big|_0^1$$

$$E(Y_2) = \frac{12}{5} - \frac{12}{7}$$

$$E(Y_2) = \frac{24}{35}$$

Exercise 9.

X_1, X_2, X_3, X_4 are IID $unif(0, 1)$

Strategy 1:

$$W_1 = \min(X_1, X_2)$$

$$P(W_1 \leq w) = 1 - P(W_1 > w) = 1 - P(X_1 > w, X_2 > w)$$

$$P(X_1 > w, X_2 > w) = P(X_1 > w)P(X_2 > w) = (1 - w)(1 - w) = (1 - w)^2$$

$$P(W_1 \leq w) = 1 - (1 - w)^2, 0 \leq w \leq 1$$

$$f_{W_1} = \frac{d}{dw}(1 - (1 - w)^2) = 2(1 - w), 0 \leq w \leq 1$$

$$E(W_1) = \int_0^1 w(2 - 2w) dw$$

$$= \int_0^1 2w - 2w^2 dw$$

$$= w^2 - \frac{2}{3}w^3 \Big|_0^1$$

$$= 1 - \frac{2}{3} = \frac{1}{3}$$

Strategy 2:

$$Y_1 = \max(X_1, X_2) \text{ and } Y_2 = \max(X_3, X_4)$$

$$W_2 = \max(Y_1, Y_2)$$

$$P(Y_1 \leq y) = P(X_1 \leq y, X_2 \leq y) = P(X_1 \leq y)P(X_2 \leq y) = y^2$$

$$f_{Y_1}(y) = \frac{d}{dy}(y^2) = 2y, 0 \leq y \leq 1$$

$$P(W_2 \leq w) = 1 - P(Y_1 > w)P(Y_2 > w)$$

$$P(W_2 \leq w) = 1 - (1 - w^2)^2$$

$$f_{W_2}(w) = \frac{d}{dw}(1 - (1 - w^2)^2) = \frac{d}{dw}(2w^2 - w^4) = 4w - 4w^3, 0 \leq w \leq 1$$

$$E(w_2) = \int_0^1 w(4w - 4w^3) dw$$

$$E(w_2) = \int_0^1 4w^2 - 4w^4 dw$$

$$= \frac{4}{3}w^3 - \frac{4}{5}w^5 \Big|_0^1$$

$$= \frac{4}{3} - \frac{4}{5}$$

$$= \frac{8}{15}$$

Strategy 3:

$$Y_3 = \text{second} - \text{largest of } \{X_1, X_2, X_3, X_4\}$$

$$W_3 = Y_3$$

$$f_{X_1, X_2, X_3, X_4}(x_1, x_2, x_3, x_4) = n! f_{X_1}(x_1) f_{X_2}(x_2) f_{X_3}(x_3) f_{X_4}(x_4) = 4! \times 1 = 24$$

$$f_{Y_3}(y) = n \binom{n-1}{2} y^2(1-y)$$

$$= 4 \binom{3}{2} y^2(1-y)$$

$$= 4 \times 3(y^2 - y^3)$$

$$12(y^2 - y^3)$$

$$= 12y^2 - 12y^3, 0 \leq y \leq 1$$

$$E(Y^3) = \int_0^1 y(12y^2 - 12y^3) dy$$

$$= \int_0^1 12y^3 - 12y^4 dy$$

$$= \frac{12}{4}y^4 - \frac{12}{5}y^5 \Big|_0^1$$

$$= 3 - \frac{12}{5}$$

$$= \frac{3}{5}$$

Exercise 10.

$$(a) P(a < X_i < b) = b - a$$

$$P(Y_1 > a, Y_n < b) = P(a < Y_i < b) = P(a < X_i < b)^n = (b - a)^n$$

$$(b) P(Y_1 > a, Y_n > b) = P(a < Y_1 < 1, b < Y_n < 1) = P(a < Y_1) - P(Y_1 > a, Y_n < b) = (1 - a)^n - (b - a)^n$$

$$(c) P(Y_1 < a, Y_n < b) = P(0 < Y_1 < a, 0 < Y_n < b) = P(Y_n < b) - P(Y_1 > a, Y_n < b) = b^n - (b - a)^n$$

$$(d) P(Y_1 < a, Y_n > b) = P(0, Y_1 < a, b < Y_n < 1) = 1 - P(Y_1 < a, Y_n > b)^c = 1 - P(a \leq Y_1) - P(b \geq Y_n) + P(Y_1 \geq a, Y_n \leq b) = 1 - P(a < Y_1) - P(b > Y_n) + P(Y_1 > a, Y_n < b) = 1 - (1 - a)^n - b^n + (b - a)^n$$

(e)

- exactly k of X_i 's are $\leq a$
- exactly $(n - k)$ of the X_i 's are $\geq b$
- no X_i is in (a, b)

$$P(Y_k < a, Y_{k+1} > b) = \binom{n}{k} a^k (1 - b)^{n-k}, 1 \leq k \leq n - 1$$

$$(f) \text{ case 1: } k + 1 \text{ between } a - b: \binom{n}{k, 1, n - k - 1} a^k (b - a)^1 b^{n-k-1}$$

$$\text{case 2: } k + 1 < a: \binom{n}{k + 1, n - k - 1} a^{k+1} (b - a)^0 b^{n-k-1} = \binom{n}{k + 1, n - k - 1} a^{k+1} b^{n-k-1}$$

$$\text{case 3: } k + 1 > b: \binom{n}{k, n - k} a^k (b - a)^0 b^{n-k} = \binom{n}{k, n - k} a^k b^{n-k}$$

$$P(Y_k, Y_{k+2} > b) = \binom{n}{k, 1, n - k - 1} a^k (b - a) b^{n-k-1} + \binom{n}{k+1, n - k - 1} a^{k+1} b^{n-k-1} + \binom{n}{k, n - k} a^k b^{n-k}$$

Exercise 11.

(a) $X_1, X_2, \dots, X_n$ - IID $\exp(\lambda)$

$$Y_1 \leq Y_2 \leq \dots \leq Y_n$$

$$W_1 = Y_1, W_2 = Y_2 - Y_1, W_3 = Y_3 - Y_2, \dots, W_n = Y_n - Y_{n-1}$$

$$Y_1 = W_1, Y_2 = W_1 + W_2, Y_3 = W_1 + W_2 + W_3, \dots, Y_n = W_1 + W_2 + \dots + W_n$$

$$f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = \lambda^n e^{-\lambda(x_1 + x_2 + \dots + x_n)}$$

$$f_{Y_1, Y_2, \dots, Y_n}(y_1, y_2, \dots, y_n) = \lambda^n e^{-\lambda(y_1 + y_2 + \dots + y_n)}$$

$$J = \left| \det \begin{pmatrix} 1 & 0 & 0 & \dots & 0 & 0 \\ 1 & 1 & 0 & \dots & 0 & 0 \\ 1 & 1 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots & \vdots \\ 1 & 1 & 1 & \dots & 1 & 1 \end{pmatrix} \right|$$

$$J = 1$$

$$f_{W_1, W_2, \dots, W_n}(w_1, w_2, \dots, w_n) = n! \lambda^n e^{-(w_1 + (w_1 + w_2) + \dots + (w_1 + w_2 + \dots + w_n))} \text{ for } 0 < w_1 < w_1 + w_2 < \dots < w_1 + w_2 + \dots + w_n$$

Exponential distribution for each of the spacings

(b) The joint PDF of $W_1, W_2, \dots, W_n$ is separable: $f_{W_1, W_2, \dots, W_n}(w_1, w_2, \dots, w_n) = f_{w_1}(w_1) f_{w_2}(w_2) \dots f_{w_n}(w_n)$

Marginal distribution: $W_1 = Y_1 \rightarrow W_1 - \text{Exp}(n\lambda)$

$W_2 = Y_2 - Y_1 \rightarrow W_2 - \text{Exp}((n-1)\lambda)$

:

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$W_i - \text{Exp}((n-i+1)\lambda)$

(c) $n = 2, X_1, X_2 - \text{Exp}(1)$

$W_1 = Y_1, W_2, Y_2 - Y_1$

$Y_1 = W_1, Y_2 = W_1 + W_2$

The random sample divides the real line into 3 pieces:

1. the interval $(0, Y_1)$ corresponding to W_1
2. the interval (Y_1, Y_2) corresponding to W_2
3. the interval Y_1, ∞ for the rest

Let the lengths of the two remaining pieces be Y_1 and $Y_2 - Y_1$

$$X = \max(Y_1, Y_2 - Y_1) = \max(W_1, W_2)$$

$$P(X \leq x) = P(\max(W_1, W_2) \leq x) = P(W_1 \leq x, W_2 \leq x) = P(W_1 \leq x)P(W_2 \leq x)$$

$$f_{Y_1, Y_2}(y_1, y_2) = 2e^{-(y_1+y_2)}, 0 \leq y_1 \leq y_2$$

$$\begin{aligned} F_x(x) &= \int_0^x \int_{y_1}^{y_1+x} 2e^{-(y_1+y_2)} dy_2 dy_1 \\ &= \int_0^x 2e^{-y_1} \int_{y_1}^{y_1+x} e^{-y_2} dy_2 dy_1 \\ &= \int_0^x 2e^{-y_1} [-e^{-y_2}]_{y_1}^{y_1+x} dy_1 \\ &= \int_0^x 2e^{-y_1} [e^{-y_1} - e^{-(y_1+x)}] dy_1 \\ &= \int_0^x 2e^{-2y_1} (1 - e^{-x}) dy_1 \\ &= (1 - e^{-x}) \int_0^x 2e^{-2y_1} dy_1 \\ &= (1 - e^{-x}) [-e^{-2y_1}]_0^x \\ &= (1 - e^{-x})(1 - e^{-x}) \end{aligned}$$

$$f_x(x) = \frac{d}{dx}((1 - e^{-x})(1 - e^{-2x}))$$

$$f_x(x) = \frac{d}{dx}(1 - e^{-2x} - e^{-x} + e^{-3x})$$

$$f_x(x) = 2e^{-2x} + e^{-x} - 3e^{-3x}$$